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Numerical simulations of blobs with ion dynamics

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Abstract

The transport of particles and energy into the scrape-off layer (SOL) region at the outboard midplane of medium-sized tokamaks, operating in low confinement mode, is investigated by applying the first-principle HESEL (hot edge-sol-electrostatic) model. HESEL is a four-field drift-fluid model including finite electron and ion temperature effects, drift wave dynamics on closed field lines, and sheath dynamics on open field lines. Particles and energy are mainly transported by intermittent blobs. Therefore, blobs have a significant influence on the corresponding profiles. The formation of a 'shoulder' in the SOL density profile can be obtained by increasing the collisionality or connection length, thus decreasing the efficiency of the SOL's ability to remove plasma. As the ion pressure has a larger perpendicular but smaller parallel dissipation rate compared to the electron pressure, ion energy is transported far into the SOL. This implies that the ion temperature in the SOL exceeds the electron temperature by a factor of 2–4 and significantly broadens the power deposition profile.

Keywords: blob dynamics, scrape-off layer transport, turbulence and flows, simulations

(Some figures may appear in colour only in the online journal)

1. Introduction

The success of particle and energy confinement in magnetically confined plasmas crucially depends on the control of transport in the outermost plasma region in contact with material surfaces. The transport regulates the particle and heat loads on the plasma-facing components, which must be controlled for a stable operation of high-power confinement devices such as ITER. An understanding of the physics governing transport at the boundary of magnetized plasmas is thus imperative.

In this paper we study the perpendicular particle and energy transport across the last closed flux surface (LCFS) using the two-dimensional model HESEL [1–4]. HESEL is an energy-conserving, four-field model based on the Braginskii equations [5] governing the dynamics of a quasi-neutral, simple plasma. It describes interchange-driven, low-frequency turbulence in a plane perpendicular to the magnetic field at the outboard midplane. In the limit of constant ion pressure the model reduces to the ESEL (edge-sol-electrostatic) model, which has successfully modelled fluctuations and profiles in JET [6], MAST [7], EAST [8], and TCV [9]. The HESEL model includes the transition from the confined region to the region of open field lines and the full development of the profiles across the LCFS. On closed field lines drift wave dynamics is included and on open field lines parallel losses from adiabatic expansion, electron heat conduction and sheath dissipation is included in the model. On both open and closed field lines we consider a single parallel length scale with the length of the unstable parallel region.

Ion dynamics on the outboard midplane of a tokamak has both experimentally [10–12] and numerically [1, 2] been shown to be a necessary, even essential, part of L–H and L–I–H transitions. Furthermore, ion dynamics is also important for power deposition on the plasma-facing components, such as the divertor. Experimentally there are only a few scrape-off layer (SOL) diagnostics that can measure electron temperature with the sampling rate in the range kilohertz to megahertz necessary to analyse the dynamics of blobs e.g. Langmuir probes and ball-pen probes [13]. Measurements of

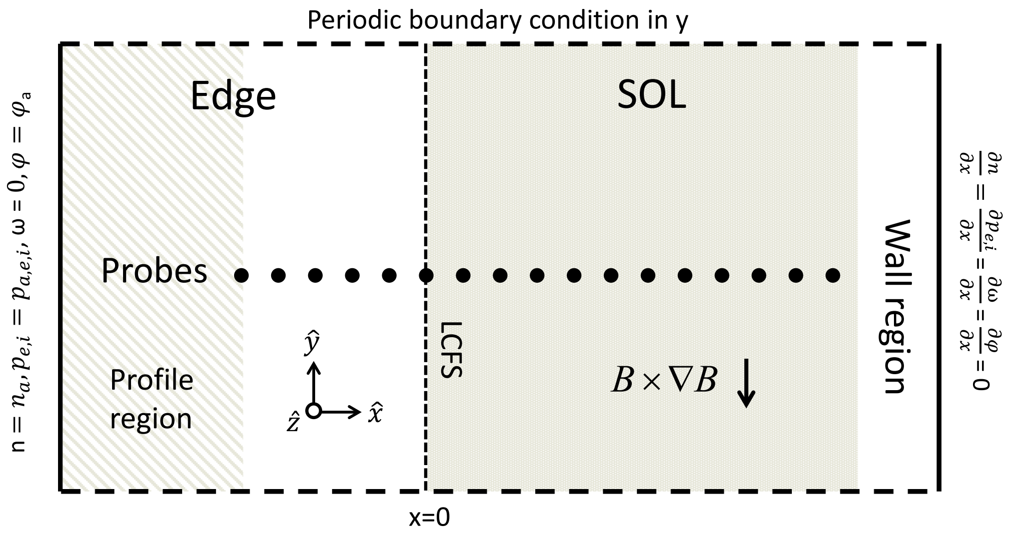

Figure 1. Setup of the simulation domain showing the closed field line region, the edge region including the inner part, where prescribed profiles are active, and the open field line region, SOL, where the parallel loss terms are active. Data time series of all fields are recorded by the probes radially separated across the edge and SOL regions, marked by black dots.

ion temperature dynamics within the same sampling rate are unfortunately even rarer. The use of retarding field analysers [14, 15] has shown that the ion temperature is typically 3–7 times the electron temperature in the far SOL [14].

The efficiency of the SOL at removing plasma can be estimated by the ratio of perpendicular to parallel particle transport, $\Gamma_\perp/\Gamma_\parallel$. If the perpendicular transport dominates, the midplane becomes disconnected from the divertor and a 'shoulder' in the density profile forms in the far SOL. The transition from parallel dominated transport to perpendicular dominated transport can be estimated by an effective collisionality parameter [16]

$$\Lambda = \frac{L_c/c_s}{1/\nu_{ei}}\frac{\omega_i}{\omega_e}, \tag{1}$$

where L_c is the connection length from midplane to outer divertor leg, c_s is the sound speed, ν_{ei} is the electron-ion collision frequency, and ω_i and ω_e are the ion and electron gyro frequencies, respectively. For $\Lambda < 1$ it is assumed that the midplane will be attached to the sheath at the divertor, and for $\Lambda > 1$ the midplane will be detached, hence the plasma can be transported far into the SOL, broadening the density profile. Experimentally, this transition has been observed at TCV [17] and more recently at ASDEX Upgrade (AUG), JET and COMPASS [18–20]. Detachment can be achieved by puffing a neutral gas into the divertor region or by recombination of plasma streaming into the divertor region. Deriving equation (1), the perpendicular transport of particles is assumed to depend on local plasma parameters. Coherent structures, blobs, are known to be responsible for this transport in the SOL on the outboard midplane. Their dynamics is not only determined by local plasma properties but also, and maybe even more importantly, by plasma conditions in the edge region, the region where they are created [21].

In this paper we will focus on blob dynamics to investigate their influence on particle and energy profiles. To characterize the structure of the blobs we will use the conditional average technique [22]. For the blobs' contributions to the radial energy fluxes we will divide the fluxes into convective, conductive and triple parts, showing that all three parts contribute to the total flux. We will investigate the influence of the divertor sheath on the perpendicular transport and the relation to shoulder formation in the SOL region. To investigate the shoulder formation in the density profile we vary the parameter Λ (see equation (1)) by changing the collisionality.

The paper is organized as follows: in section 2 we present the HESEL model, in section 3 we present the result and finally in section 4 we present our conclusions.

2. Model equations

In this section we present the model equations applied in the present work; for a more detailed description of the model see [4]. We base our description on the interchange instability of a non-uniformly magnetized plasma on the outboard midplane of a medium-sized tokamak. In the case of an unstable equilibrium, the drive of the fluctuating motions is a convective transport of particles and energy outwards along the major radius of the toroidally magnetized plasma. A self-consistent description of this collective dynamics is here considered as a mechanism for intermittent turbulence in the plasma boundary region. In order to perform long simulations producing sufficient data to allow detailed statistical analysis, we reduce our description to two spatial dimensions and parallel dynamics along open magnetic field lines in the SOL is parametrized. The model is solved in a local slab geometry, $(\hat{x}, \hat{y}, \hat{z})$, with the unit vector $\hat{z}$ along the inhomogeneous toroidal magnetic field. The magnetic field is approximated by $B(r) = B_0(R + a)/(R + a + x)$, where R and a are the major and minor radius of the toroidal plasma and B_0 is the magnetic field strength at the outboard midplane. Figure 1 shows a sketch of the setup.

We use the so-called Bohm normalization defined by

$$\omega_{i0}t \to t,\ \frac{x}{\rho_s} \to x,\ \frac{T_{e,i}}{T_{e0}} \to T_{e,i},\ \frac{e\phi}{T_{e0}} \to \phi,\ \frac{n}{n_0} \to n, \quad (2)$$

where $\omega_{i0} = eB_0/m_i$ is the ion gyro frequency at magnetic field strength at the outboard midplane, B_0, $\rho_s = \sqrt{T_{e0}/(m_i\omega_{i0}^2)}$ is the cold-ion hybrid thermal gyro-radius and n_0 and T_{e0} are characteristic values of the particle density and electron temperature, respectively.

The model equations read [4]:

$$\frac{d}{dt}n + n\mathcal{K}(\phi) - \mathcal{K}(p_e) = \Lambda_n, \quad (3)$$

$$\frac{d^0}{dt}w^* + \{\nabla\phi, \nabla p_i\} - \mathcal{K}(p_e + p_i) = \Lambda_{w^*}, \quad (4)$$

$$\frac{3}{2}\frac{d}{dt}p_e + \frac{5}{2}p_e\mathcal{K}(\phi) - \frac{5}{2}\mathcal{K}\left(\frac{p_e^2}{n}\right) = \Lambda_{p_e}, \quad (5)$$

$$\frac{3}{2}\frac{d}{dt}p_i + \frac{5}{2}p_i\mathcal{K}(\phi) + \frac{5}{2}\mathcal{K}\left(\frac{p_i^2}{n}\right) - p_i\mathcal{K}(p_e + p_i) = \Lambda_{p_i}, \quad (6)$$

where n is the particle density, $w^* = \nabla^2\phi + \nabla^2 p_i$ is the generalized vorticity, ϕ is the electrostatic potential, and p_e and p_i are the electron and ion pressure, respectively. Temperatures are defined by $T_{i,e} = p_{i,e}/n$. The material derivative is defined as $\frac{d}{dt} = \frac{\partial}{\partial t} + \frac{1}{B(x)}\hat{z} \times \nabla\phi \cdot \nabla$, except in the vorticity equation (equation (4)), where the magnetic field is taken as constant, B_0. The curvature operator is given as $\mathcal{K} = \nabla\left(\frac{1}{B(x)}\right) \cdot \hat{z} \times \nabla$.

The generalized vorticity is the manifestation of the polarization current in the model and describes charge separation due to the inertia in the ion response to changes in the $\boldsymbol{E} \times \boldsymbol{B}$ and diamagnetic drifts.

The right-hand sides of equations (3)–(6) represent perpendicular resistivity, perpendicular electron and ion energy conduction, ion viscosity, and parametrized parallel dynamics [4]:

$$\Lambda_n = D_e\nabla_\perp^2 n - \frac{n}{\tau} - \alpha\left(\tilde{T}_e + \frac{\bar{T}_e}{\bar{n}}\tilde{n} - \tilde{\phi}\right) - \frac{n - n_p}{\tau_p}, \quad (7)$$

$$\Lambda_{w^*} = D_i\nabla_\perp^2 w^* - \frac{w^*}{\tau} + \frac{\rho_s}{L_c}\left[1 - \exp\left(\phi_m - \frac{\phi_s}{T_{e,s}}\right)\right] - \alpha\left(\tilde{T}_e + \frac{\bar{T}_e}{\bar{n}}\tilde{n} - \tilde{\phi}\right), \quad (8)$$

$$\Lambda_{p_e} = \frac{5}{2}D_e\nabla_\perp^2 p_e + \left(\frac{16}{6} - \frac{5}{2}\right)\nabla\cdot(n\nabla_\perp T_e) - \frac{9}{2}\frac{p_e}{\tau} - \frac{T_e}{\tau^{SH}} - \alpha\bar{T}_e\left(\tilde{T}_e + \frac{\bar{T}_e}{\bar{n}}\tilde{n} - \tilde{\phi}\right) - \Theta - \frac{p_e - p_{e,p}}{\tau_p}, \quad (9)$$

$$\Lambda_{p_i} = D_i\nabla_\perp^2 p_i - D_i T_i\nabla_\perp^2 n - \frac{9}{2}\frac{p_i}{\tau} + \Theta - \frac{p_i - p_{i,p}}{\tau_p} + p_i\Lambda_{w^*}, \quad (10)$$

where the normalized neoclassic transport coefficients are given by

$$D_i = \left(1 + \frac{R}{a}q^2\right)\frac{\rho_{i0}^2\nu_{ii,0}}{\rho_{s0}^2\omega_{i0}},$$
$$D_e = \left(1 + \frac{T_{i0}}{T_{e0}}\right)\left(1 + \frac{R}{a}q^2\right)\frac{\rho_{e0}^2\nu_{ei,0}}{\rho_{s0}^2\omega_{i0}}, \quad (11)$$

as derived in [4] and [6]. q is the safety factor and is in this model taken to be q_{95}. ρ_{e0} and ρ_{i0} are the electron and ion gyro radii. The term $\Theta = \frac{3m_e}{m_i}\nu_{ei}(p_e - p_i)$ in equations (9)–(10) is the energy transfer between the electron and ion channels. We define the mean and fluctuating components as

$$\bar{f} = \frac{1}{L_y}\int_0^{L_y} f dy,\ \tilde{f} = f - \bar{f}. \quad (12)$$

The terms involving parametrization of the parallel dynamics will be defined in the following subsection.

At the inner part of the edge region the density and pressure profiles are forced towards prescribed profiles n_p, $p_{e,p}$ and $p_{i,p}$ with a time scale of τ_p (see figure 1). These profiles resemble typical profiles in the edge region of toroidally magnetized plasmas and act as sources of particles and energy, but also remove 'unphysical' instabilities otherwise generated in the vicinity of the inner boundary of the numerical system; a boundary that is of course not present in a tokamak.

For the density, pressures, generalized vorticity and generalized potential, $\phi^* = \phi + p_i$, we have used Dirichlet boundary conditions at the inner radial boundary and zero Neumann boundary conditions at the outer radial boundary. The Dirichlet conditions are in agreement with the prescribed profiles. All fields are periodic in the y direction. The set of equations is solved by employing a finite difference scheme with a symmetric, energy and enstrophy conserving discretization of the nonlinear advection terms and an explicit third-order stiffly stable time integrator with diffusive terms treated implicitly using operator splitting [23].

2.1. Parametrization of the parallel dynamics

The ballooning character of the radial transport in the edge region implies that the corresponding SOL region plasma source is concentrated to a region that extends 30 degrees above and below the outboard midplane [24]. In that respect we define a parallel ballooning length corresponding to the poloidal angle of this unstable region as $L_b = qR$ (see also [25, 26]). Using characteristic parameters from medium-sized tokamaks like EAST and AUG we obtain $L_b \sim 8$ m and this length should be compared to the connection length from outboard midplane to outer divertor leg along the magnetic field lines, $L_c \sim 15$ m. As the plasma propagates radially outward it will expand along the magnetic field lines with a speed of $2M_{\|}c_s \sim 50\,000$ m s^{-1} where the hot ion sound speed, $c_s = \sqrt{(T_e + T_i)/m_i}$ has been calculated using $T_i = T_e = 25$ eV and we have set the parallel Mach number to $M_{\|} = 0.5$. As the radial velocity of a blob typically will be a few percent of the sound speed, we assume that the time a blob needs to cross the SOL is smaller than the time the plasma from the outboard midplane takes to reach the divertor along the magnetic field lines; $V_b/L_{SOL} > c_s/L_c$. As the blob expands parallel we will thus observe a decrease of the particle density, electron and ion pressures and the generalized vorticity due to sonic parallel expansion without the influence of the divertor.

Losses due to advection and electron heat conduction along the magnetic field lines in the SOL region are represented by the damping rates,

$$\frac{1}{\tau} = \frac{2M_{\|}c_s}{L_b}, \quad \frac{1}{\tau^{SH}} = \frac{X^{SH}_{\|,e}}{L_b^2}, \tag{13}$$

where the Spitzer-Härm (SH) expression for the electron heat diffusivity is given by $X^{SH}_{\|,e} = 3.16V_e^2/\nu_{ee}$. Both the hot ion sound speed, c_s, the electron–electron collision frequency, ν_{ee}, and the electron thermal speed, V_e, are evaluated using the local value of electron and ion temperatures.

The compression of the parallel current is approximated on open field lines by a sheath dissipation term entering the vorticity in equation (8) where $\phi_m = \ln(\sqrt{m_i/2\pi m_e})$ is the Bohm potential. Using the poloidal average defined in equation (12) as an approximation for the flux surface average, the connected divertor condition is modelled by using the full fields and the shielded or disconnected divertor condition is modelled by using only the mean fields, i.e.

$$\begin{aligned} \text{connected:} \quad \phi_s &= \phi(x, y, t),\ T_{e,s} = T_e(x, y, t) \\ \text{disconnected:} \quad \phi_s &= \bar{\phi}(x, t),\ T_{e,s} = \bar{T}_e(x, t). \end{aligned}$$

On closed field lines the compression of the parallel current is approximated by drift wave terms, i.e. the terms in equations (7)–(9) involving α. The coefficient is given by (see [25])

$$\alpha = \frac{2T_{e0}}{\nu_{ei} m_e L_{\|}^2 \omega_{i0}}, \tag{14}$$

where the parallel length scale is, unless otherwise stated, chosen as the parallel ballooning length, $L_{\|} = L_b = qR$. In deriving equation (14), we have assumed that the mean components of density, potential and electron temperature do not have any parallel variation, $\nabla_{\|}\bar{n} \sim \nabla_{\|}\bar{T}_n \sim \nabla_{\|}\bar{\phi} \sim 0$, and that second-order terms (like $\tilde{n}\tilde{T}_e$) can be neglected. We note that this representation is a significant simplification of the parallel dynamics on closed field lines and that it is quite sensitive to the choice of parallel length and reference electron temperature.

3. Numerical results

Typical parameters for HESEL simulations are parameters relevant for a medium-sized tokamak, such as EAST or AUG.

Unless otherwise stated, the reference parameters used in this paper are: magnetic field strength $B_0 = 1.9$ T, $q = q_{95} = 5.32$, major radius $R = 1.65$ m, minor radius $a = 0.5$ m, and parallel connection length to divertor $L_c = 15.0$ m. In the wall shadow region the parallel connection length is reduced to 2.5 m. This region thus acts as a damping layer which also prevent reflections from the outer wall. Subscript 0 refers to the expected value in the vicinity of the LCFS. In the low-collisionality case, we use density $n_0 = 1.5 \times 10^{19}$ m^{-3}, electron temperature $T_{e0} = 25$ eV, and ion temperature $T_{i0} = 25$ eV. With these parameters we obtain the following values for the Bohm normalized transport coefficients: $D_e = 0.0043$, $D_i = 0.073$, $1/\tau_0 = 0.000016$, $1/\tau_0^{SH} = 0.00098$ (the subscripts 0 refer to the damping rates calculated using the electron reference temperature, T_{e0}), $\alpha = 0.000274$ and $\rho_s = 0.38$ mm. The effective collisionality parameter is $\Lambda_0 = 0.758$ for these parameters. A connected divertor condition is used for this set of parameters. In the high-collisionality case, we use density $n_0 = 3.5 \times 10^{19}$ m^{-3}, electron temperature $T_{e0} = 15$ eV, and ion temperature $T_{i0} = 15$ eV. The Bohm normalized transport coefficients are: $D_e = 0.02$, $D_i = 0.34$, $1/\tau_0 = 0.012$, $1/\tau_0^{SH} = 0.02$, $\alpha = 0.000036$ and $\rho_s = 0.29$ mm. The effective collisionality parameter is $\Lambda_0 = 4.6$ for these parameters. We have performed two simulations with these parameters; one with a connected SOL divertor condition, and one with disconnected condition. In all the cases considered the wall region is connected.

For the numerical domain see figure 1. We have used $L_x = L_y = 95.0$ mm with an edge width of $L_{edge} = 34.4$ mm, a SOL width of $L_{SOL} = 47.5$ mm and a wall shadow region of $L_{wall} = 12.3$ mm. The resolution was chosen to be $N_x = N_y = 1024$.

3.1. Blob evolution

In figures 2–4 we show snapshots of all the relevant fields from three HESEL simulations using the above parameters. In the low-collisionality case (figure 2) blobs being ejected into the SOL quickly lose their momentum, and are almost stopped approximately 2 cm from the LCFS. In the high-collisionality case (figure 3) larger blobs are observed, and they penetrate further into the SOL than for lower collisionality. Finally, for a disconnected divertor condition in the SOL (figure 4) there is no sink for the momentum of the fluctuations and powerful blobs can be observed completely penetrating the SOL and entering the wall region, where they are damped due to strong parallel damping.

Figure 5 show the time evolution from one of the probes in the simulation shown in figure 2. The probe is positioned in the SOL just outside the LCFS, indicated by a black dot on the density, temperatures and potential plots in figure 2. The intermittent nature of the system is

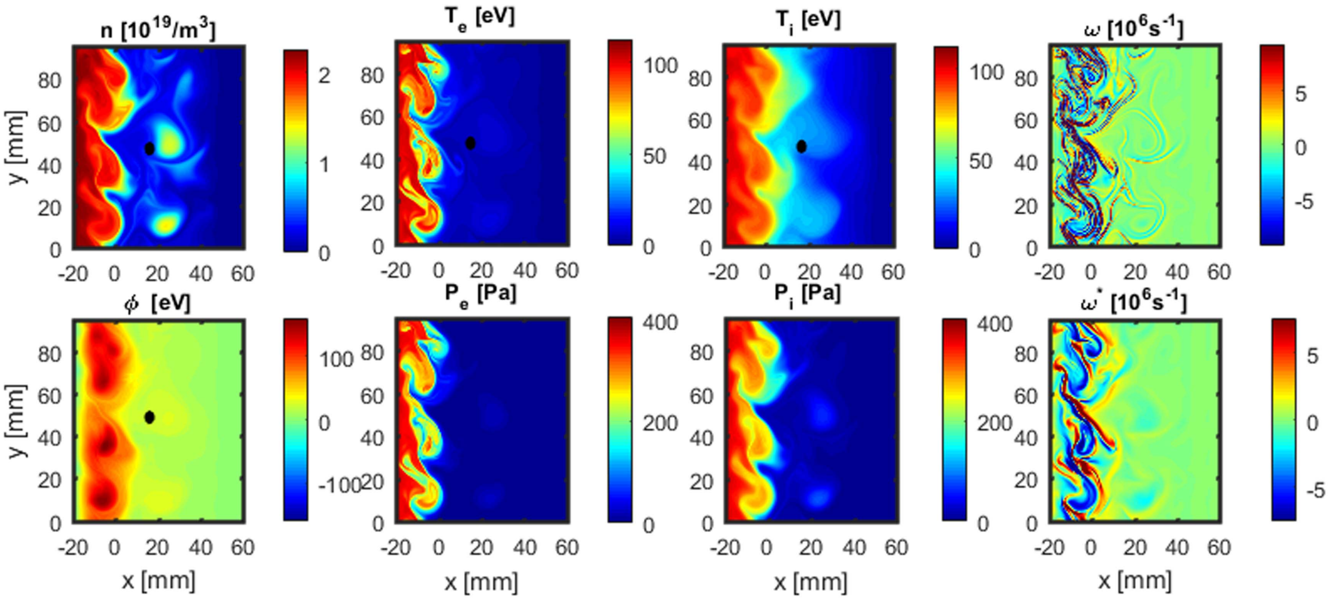

Figure 2. Snapshot of a HESEL simulation using connected divertor condition $\Lambda_0 = 0.75$ and for $t = 0.791$ ms. Frames displayed are: density, n, electron temperatures T_e, ion temperature Ti, vorticity w, electric potential ϕ, electron pressure p_e, ion pressure p_i and generalized vorticity $w*$. The black dot is the probe used for the time series in figure 5.

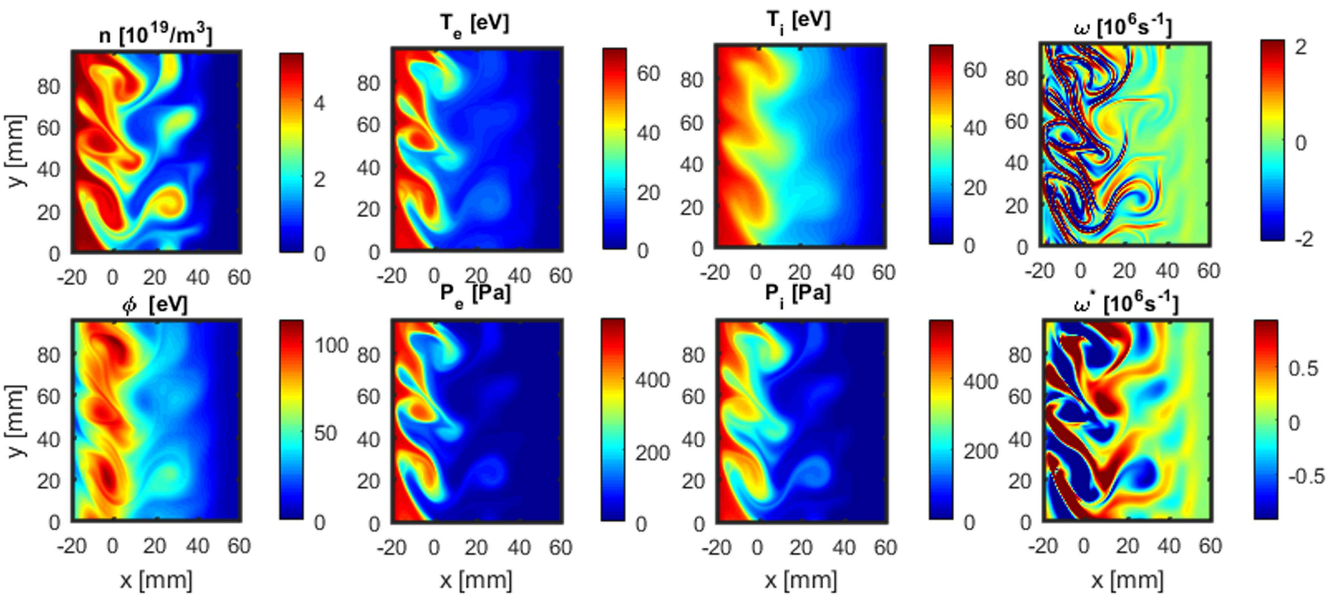

Figure 3. As for figure 2 but $\Lambda_0 = 4.6$.

clearly seen in all the plots. For the turbulent flux signals, $\Gamma_n = nv_x$, $\Gamma_{P_\mathrm{e}} = nT_\mathrm{e} v_x$, $\Gamma_{P_\mathrm{i}} = nT_\mathrm{i} v_x$, all the radial transport takes place in 3−4 localized events during the considered time window. We also observe that the ion temperature and energy flux are significantly larger than the electron temperature and energy flux. The electric potential variation follows density and temperature variations closely.

3.2. Drift wave dynamics

In previous papers where the HESEL model was used to investigate the L–I–H transition [1, 2], the model only included the interchange mode (IM) modified by finite ion temperature effects and dissipation terms. This paper accounts for the influence of finite parallel wavelength and includes the resistive ballooning mode (RBM) [27]. The maximum linear growth rate in the IM peaks at a low poloidal wave number and an instability with a wavelength of the width of the numerical box would typically be generated. For the RBM the linear growth rate increases from 0 with increasing k_θ to a maximum and a larger poloidal wave number will be expected to dominate.

In figure 6 we show snapshots of the density from individual simulations each at the time, where the perturbation of the initial unstable plasma becomes visible. An inspection of equation (14) reveals that $\alpha \propto T_{\mathrm{e},0}^{2.5} / (n_0 L_\parallel^{\,2})$. The figure displays a variation of the parallel length scale $L_\parallel$, safety factor q,

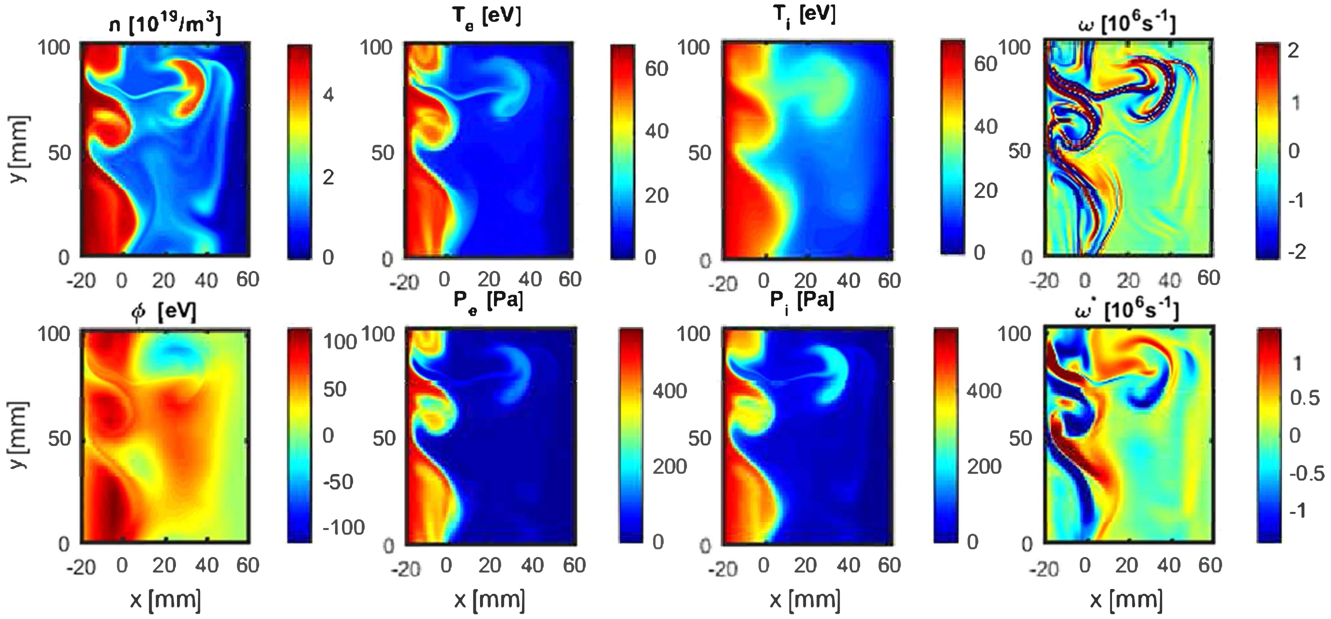

Figure 4. As for figure 2 but $\Lambda_0 = 4.6$ and disconnected divertor condition.

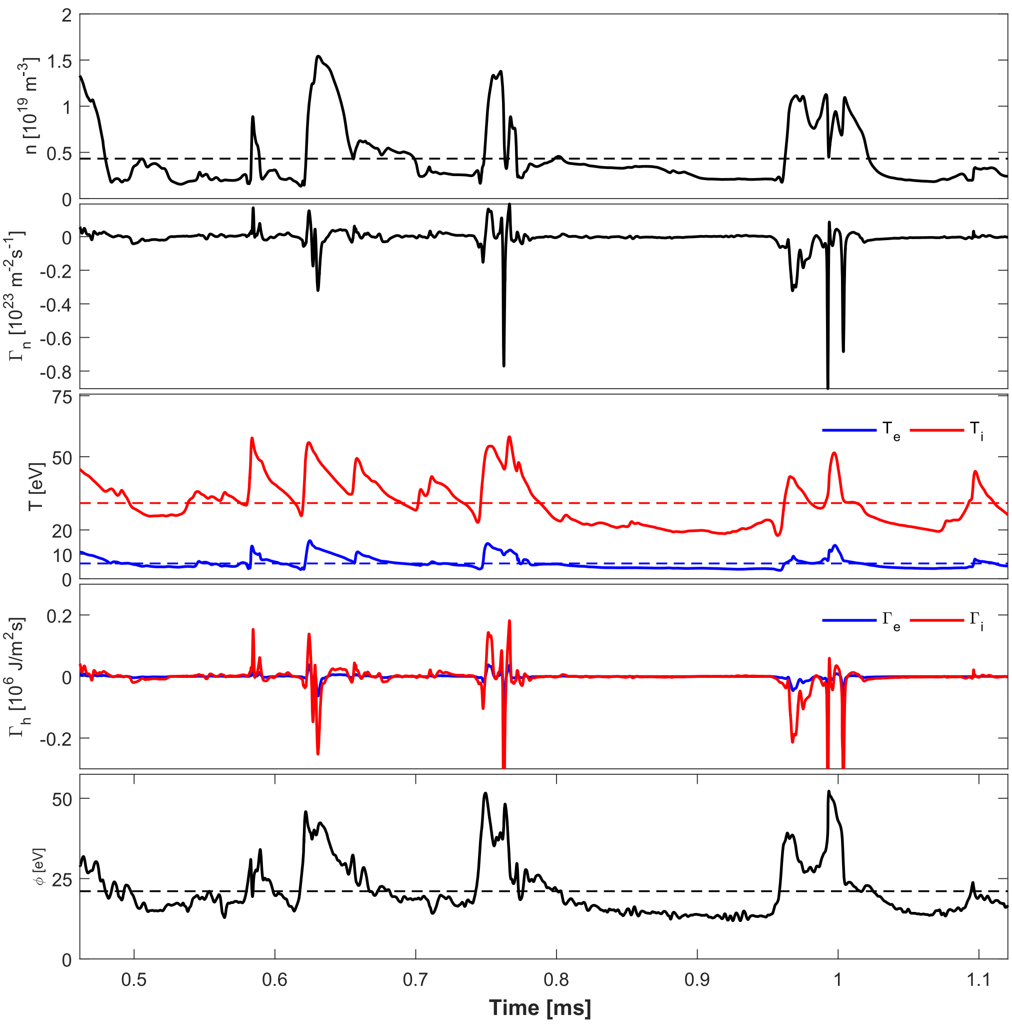

Figure 5. Time evolution of the density, particle flux, temperature, energy flux and electric potential at a radial position $x = 18.0$ mm (see figure 2). Red curves correspond to ion dynamics and blue to the electrons.

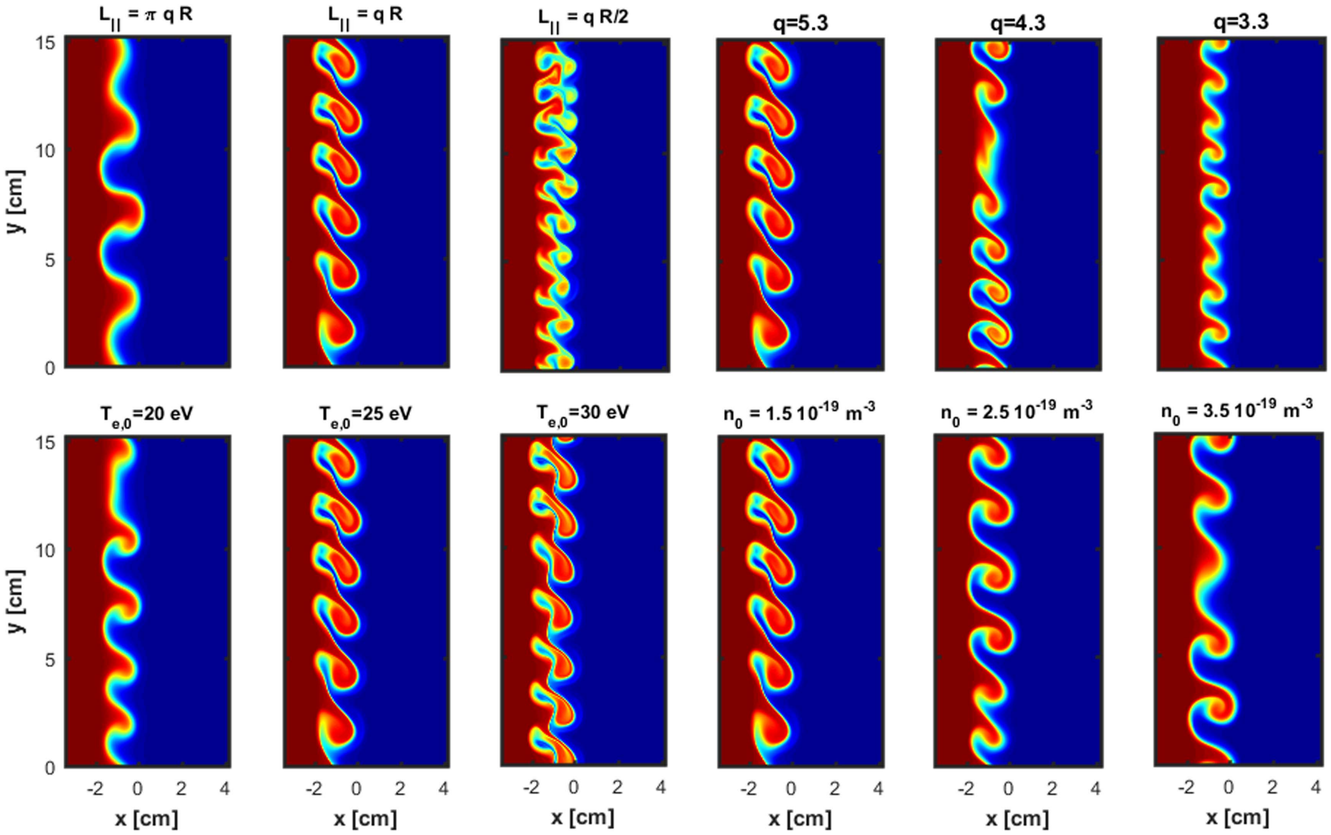

Figure 6. Snapshots of the density at the time where the initial perturbation of the unstable plasma become visible. Each frame is an individual simulation. Input parameters are as for figure 2 except that $L_y = 15$ cm and the input parameter is given in the titles. Red areas denote high densities and blue areas denote low densities.

reference density n_0 and electron temperature $T_{e,0}$. All the simulations are for the $\Lambda_0 = 0.75$ case, except for the parameter given in the title of each frame. The first three frames show a variation of the parameter $L_{\|}$. The system is very sensitive to change in this parameter, and an increase results in a strong instability. In a full simulation these instabilities reflect the size of the blobs in the SOL. Note that a change in $L_{\|}$ will only change α. As mentioned in section 2.1 we have made the somewhat natural choice of setting the parallel length equal to the ballooning length, $L_{\|} = qR$, linking this length to the safety factor q. The next three frames in figure 6 are for a variation in q and we overall observe similar changes as for changing $L_{\|}$. We have also considered a variation of the reference density n_0 and the reference electron temperature $T_{e,0}$. Again, as α is decreased larger spatial instabilities are observed.

To gain a closer insight into the influence of the drift wave term we have derived the cross power spectral density for the fluctuations. For two temporal signals, $h(t)$ and $g(t)$, obtained from the same numerical probe (see figure 1), the cross power spectral density is derived by Fourier transforming the cross-correlation function, $R_{h,g}(\tau) = \langle h(t) g(t + \tau) \rangle$. The complex cross power spectrum may be written as [28]

$$S(f) = |S(f)| \exp(i\theta(f)). \tag{15}$$

The quantities $|S|$ and θ are referred to as the amplitude and the cross phase spectrum, respectively. The term $\theta(f)$ directly gives the phase difference between h and g at the given frequency f. For the turbulent particle flux, $\Gamma(x, t) = h(x, t) g(x, t) = n(x, t) v_x(x, t)$ we plot in figure 7 $|S|$ and θ for two different radial positions; $x_1 = -4$ mm and $x_2 = +8$ mm. For both positions we observe a relatively broad spectrum up to approximately 70 kHz, consistent with blobs of diameters of $\delta \sim 1 - 2$ cm and with velocities of $v_b \sim 500 - 1.000$ m s^{-1}. The phase, θ, shows large fluctuations, but we generally observe negative phases inside the LCFS and likewise positive phases outside the LCFS, as seen in figure 7. We notice that even with the change of sign the calculated phases are nearly all within $\pm\pi/2$, which corresponds to a positive radial particle transport. Maximum transport is obtained for $\theta = 0$.

In figure 8 we display the radial profiles of $\langle\theta\rangle$ for a combination of density, electron and ion temperature signals versus the radial velocity signal. $\langle\theta\rangle$ is calculated as a weighted average over all frequencies: $\langle\theta\rangle = \sum_k \theta_k S_k / Z$, where $Z = \sum_k S_k$. In the edge region, where the drift wave term is active, we generally observe a negative weighted phase difference for all cases. In the SOL region we likewise observe a moderate positive weighted phase. In the far SOL, $x > 25$ mm, for $\Lambda_0 = 0.75$, the weighted phase becomes approximately $\pi/2$, as most blobs in this case do not propagate into the far SOL. We note that we are not obtaining a phase difference near $\pm\pi/2$ as expected numerically for pure drift wave turbulence in the edge region, well inside the

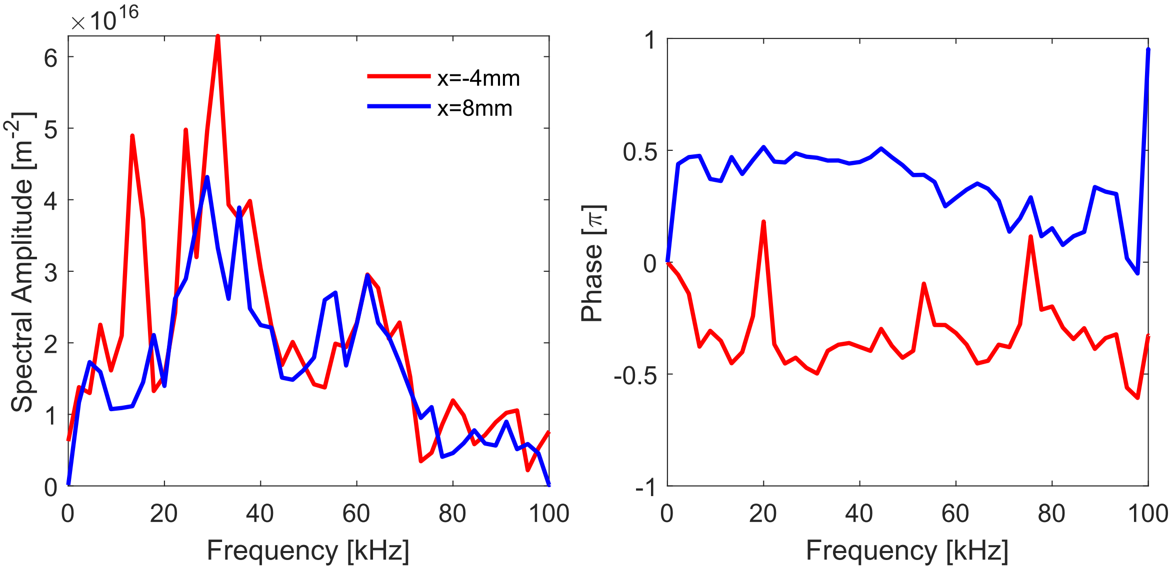

Figure 7. The amplitude $S(f)$ and phase $\theta(f)$ of the radial particle flux for two radial positions obtained from the cross power spectral density (see equation (15)). Data are from the HESEL simulations presented in figure 2.

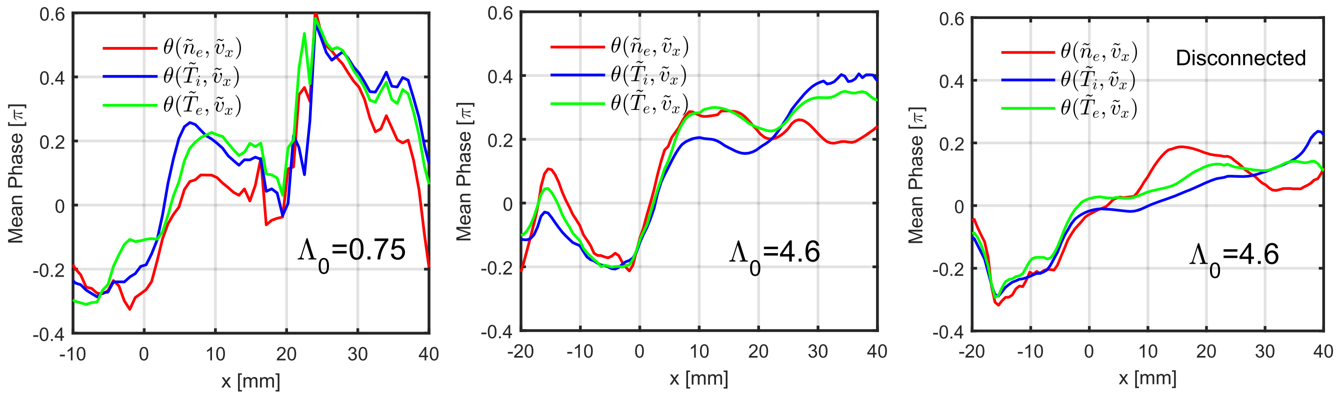

Figure 8. Radial profiles of the average phase difference calculated from the cross power spectral density. Data are from the HESEL simulations presented in figures 2–4.

LCFS. This observation is somewhat in agreement with [29], where the presence of both drift wave and interchange dynamics make it impossible to separate the two effects.

3.3. Conditional average technique

The basic idea of a conditional analysis is discussed by, e.g., Johnsen *et al* [22]. Two time records, A and B, of correlated, randomly varying signals are considered. One, say A, is used as a reference. For A we determine its maximum value, $A(t_*)$. If this value fulfils the prescribed condition, $(A(t_*) - \bar{A}) > \beta\sigma_A$ with $\bar{A}$ and σ_A being the mean and the RMS of A, respectively, a sub-series is selected from signal B in the time interval $\{t_* - \tau; t_* + \tau\}$. Signal A is then zero-padded in a time interval $\{t_* - 2\tau; t_* + 2\tau\}$ and the procedure is repeated as long as the condition on A is fulfilled. The time τ is usually taken to be of the order of the correlation time for the signals. The selected time sequences will therefore always be non-overlapping. The resulting sets of sub-series are considered as independent conditional realizations for the ensuing statistical analysis. This technique is widely used in analysing experimentally obtained data from, e.g., Langmuir probes [28, 30–34].

In figure 9 we show the results for $\Lambda_0 = 0.75$ of the conditional averaging analysis applied to the time signals from the probes positioned radially in the SOL region and using the density signal as reference signal, A. Using the density signal also as signal B (figure 9(a)), we observe a well-defined blob, which decreases in amplitude as it propagates across the SOL. As the blob reaches the middle of the SOL a significant broadening of the conditional signal is observed, consistent with the fact that the radial velocity of the blob significantly decreases around this radial position. The blob will therefore slowly decay and in this process it will expand due to perpendicular diffusion. Using one of the two temperatures as signal B (figure 9(b)–(d)), we observed that the amplitudes and widths of the conditional average signals decreased significantly. As the electron heat conduction is much faster than the parallel advection, $1/\tau \ll 1/\tau^{SH}$, the electron temperature has nearly returned to the background level when the blob reaches the middle of the SOL, with a perturbation of $\Delta T_e \sim 5$ eV. The ion temperature

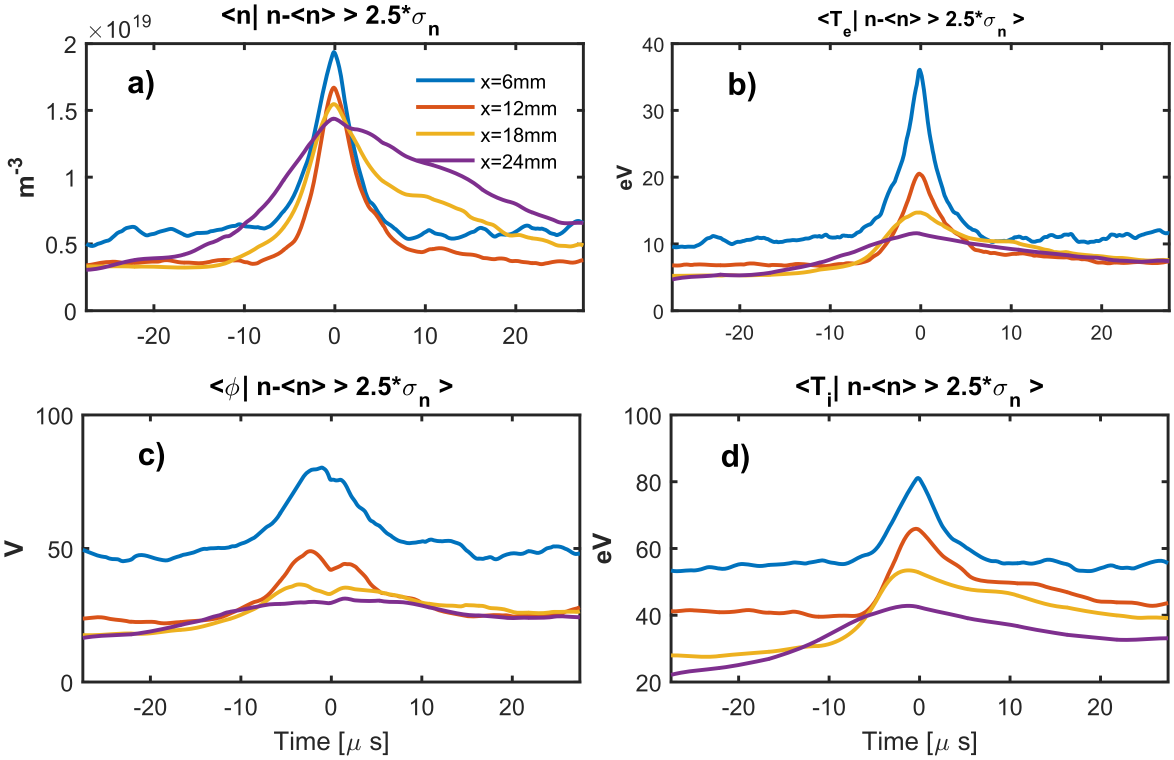

Figure 9. Conditional average wave forms of density, temperature and electric potential using the density as the reference signal. Data taken from the HESEL simulation presented in figure 2.

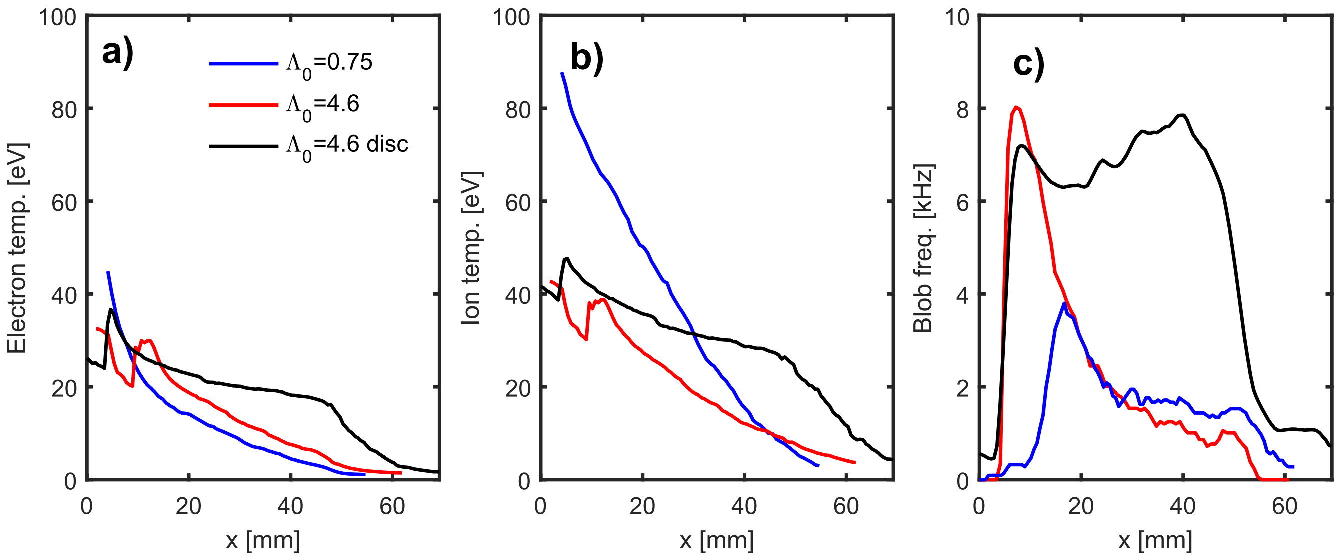

Figure 10. (a) Radial electron temperature profiles and (b) radial ion temperature profile of the maximum value of the conditional averaged blob for the three cases: $\Lambda_0 = 0.74$ (full lines), $\Lambda_0 = 4.6$ (dashed lines) and $\Lambda_0 = 4.6$ for disconnected divertor (dotted lines). In (c) the corresponding radial profile of frequency of events is shown.

perturbation, on the other hand, is still large, $\Delta T_i \sim 20$ eV. We also observe that the ion temperature profile is broader than the electron temperature profile, consistent with $D_e \ll D_i$.

Applying the conditional averaging technique to the electrical potential signal (figure 9(c)) reveals a signal somewhat similar to the electron temperature signal. The potential is linked to the electron temperature by the sheath term in the vorticity equation (equation (4)) and we roughly observe from figures 9(b)–(c) that $\phi \sim \phi_m T_e$. Another notable feature of the potential signal is that the two polarities of the potential associated with a dipole are not observed here.

In figures 10(a)–(b) we show the profiles of the maximum temperature of the conditional averaged blob in figures 9(b) and (d). For $\Lambda_0 = 0.75$ we observed an ion temperature that was twice as high as the electron temperature as the blob entered the SOL. This ratio increases to 3−4 for $x > 15$ mm. This result is consistent with experimental

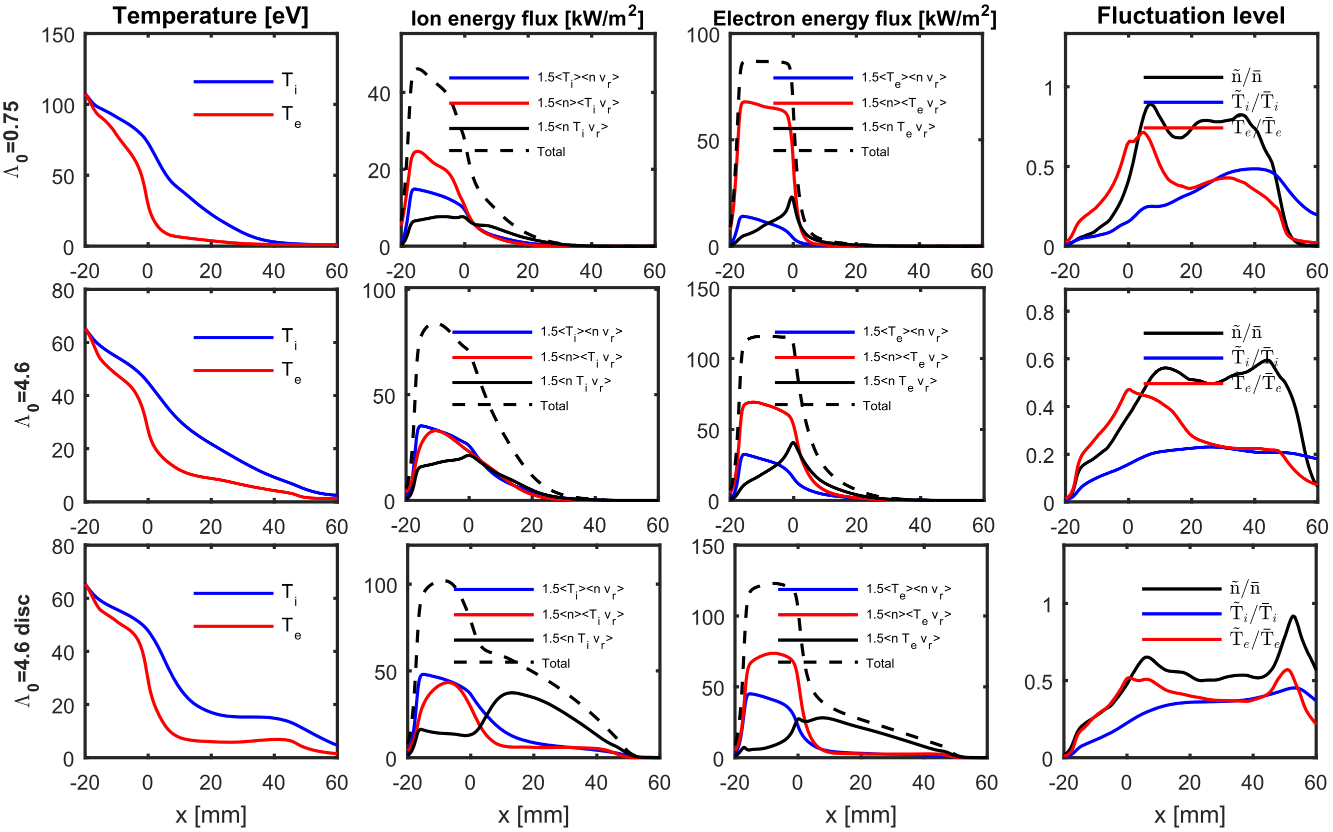

Figure 11. Radial profiles of temperatures (column 1 in the figure); ion and electron turbulent energy fluxes decomposed into convective (red), conductive (blue), triple product (black) and total flux (dashed) (columns 2–3); and relative fluctuation levels (column 4). The top row is for $\Lambda_0 = 0.75$, the middle row is for $\Lambda_0 = 4.6$ and the bottom row is for $\Lambda_0 = 4.6$ and disconnected divertor conditions.

measurements using retarding field analysers [14]. For $\Lambda = 4.6$ and the connected divertor condition we observed that the ratio between electron and ion temperature is significantly reduced compared to the $\Lambda_0 = 0.75$ case. In figure 10(c) we plot the radial profile of the frequency of blobs measured by the conditional average method. Using the condition $n - \bar{n} > 2.5\sigma_n$ results in very few blobs just outside the LCFS. For the disconnected divertor condition we observed a temperature ratio which is just above 1 in the SOL and also a nearly constant frequency of blobs going all the way to the wall region. We note that these simulations use the same reference temperatures, $T_{e0} = T_{i0}$, as well as forced pressure profiles, $p_{e,p} = p_{i,p}$, in the inner edge region.

3.4. Radial profiles

In figure 11 we display the profiles of the temperatures, turbulent energy fluxes, and fluctuation levels. The different rows are for the three simulations in question. All the profiles have been averaged both poloidally and in time. The electron temperature profiles have sharp gradients close to the LCFS approaching a low value already within 1 cm of the LCFS, whereas the ion temperature profiles are much broader with a width of $\sim 2 - 4$ cm. We decompose the turbulent ion and electron energy fluxes, $\langle n_e T_\alpha \tilde{v}_r \rangle$, $\alpha = \{i, e\}$ and $\langle\rangle$ define time and poloidal average into a conductive energy flux, $\langle n_e \rangle \langle \tilde{T}_\alpha \tilde{v}_r \rangle$, a convective energy flux, $\langle T_\alpha \rangle \langle \tilde{n} \tilde{v}_r \rangle$, and a pure nonlinear flux, the so-called triple product, $\langle \tilde{n}_e \tilde{T}_\alpha \tilde{v}_r \rangle$. In the edge region the energy fluxes are dominated by the conductive and convective parts, but in the SOL the largest contributions come from the triple products. This is especially the case for the disconnected divertor condition, a consequence of the very intermittent dynamics in the SOL for this case. The fluctuation levels, defined here as the RMS value divided by the mean value, can be observed to be high throughout the SOL and with a density and electron temperature significantly higher than the ion temperature in the inner part of the SOL.

In figure 12 we show the radial profiles of the density, turbulent particle fluxes $\Gamma_\perp$ and poloidal velocities for the three cases in question. As Λ_0 is increased from 0.75 to 4.6 the large gradient just outside the LCFS is replaced by a much broader density profile, characteristic of a so-called shoulder profile (see [18, 19]). Also the perpendicular particle flux profile gets broader as Λ increases, which demonstrates that the effective area, or wetted area, of the divertor increases. For a disconnected divertor the generated blobs can cross the SOL completely and enter the wall region. The SOL is a transit region and we observe nearly constant density profiles. The perpendicular particle flux at the entrance to the wall region is approximately 30% of the LCFS values. From figure 11 we observed that approximately 20% of the ion energy and 10% of the electron energy reach the wall region in the disconnected case, displaying that the ability of the SO to remove plasma is reduced in this case.

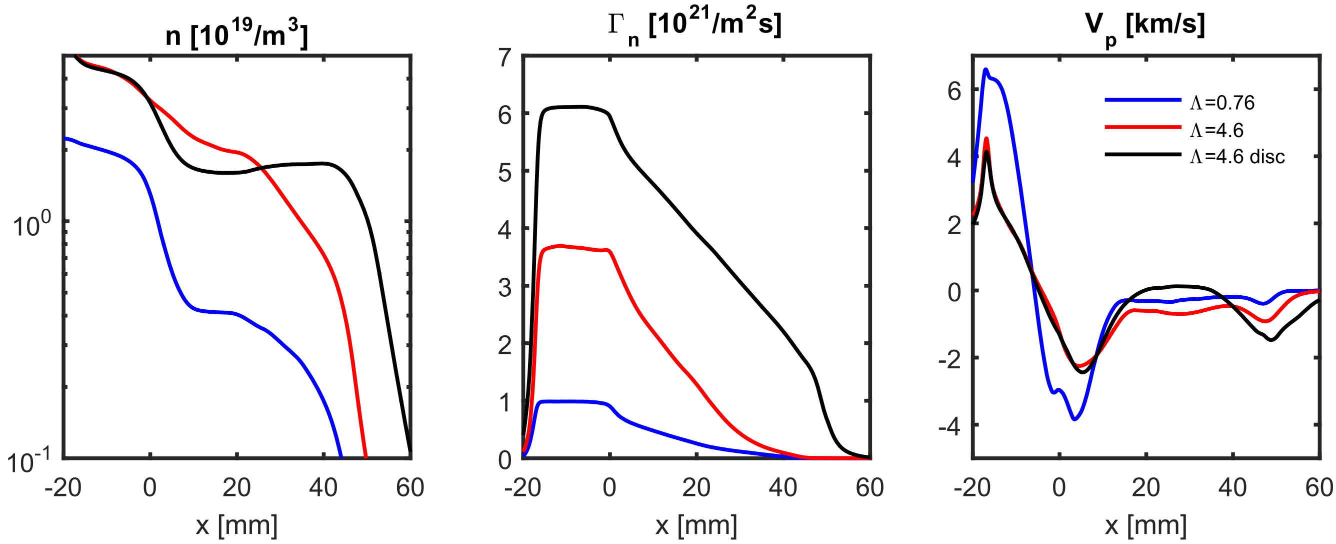

Figure 12. Radial profiles of density, turbulent perpendicular particle flux Γ_n, and the poloidal velocity in the edge and SOL for the three different simulations.

The poloidal velocities show a strong shear across the LCFS with a large negative velocity just 5 mm outside the LCFS. This position is also the position where the weighted phase changes sign (see figure 8). We also note that the poloidal velocity is apparently unaffected by the choice of divertor condition, as the profiles for $\Lambda_0 = 4.6$ with connected/disconnected divertor condition are nearly identical. The poloidal velocity is negative in all cases on open field lines and is thus in the ion diamagnetic direction. Finally, we should note that a similar poloidal flow structure in the region around the LCFS is often observed in experiment with the same flow direction and comparable flow speed (see e.g. [35]).

4. Conclusions

We have examined blob propagation across the SOL due to interchange dynamics on the outboard midplane of a tokamak using the 2D HESEL model. We have shown that blobs have a high ion temperature but low electron temperature perturbation, in agreement with experimental observations [13, 14]. This appears to be the first time that the experimentally observed differences between the temperatures in the SOL have been modelled. Energy losses to plasma-facing components, such as the divertor, thus depend heavily on the ion temperature dynamics.

The parallel dynamics in HESEL is parametrized. Transport of plasma across the LCFS is known to be concentrated in a region of 60 degrees on the outboard midplane [24], and we have used this to define a flux tube with a parallel ballooning length of $L_b = qR$. This length is used to parametrize parallel losses due to advection along the magnetic field lines. We are aware that parallel electron heat conduction is a relatively fast process, and an equilibrium state stretching from the outboard midplane to the outer divertor establishes quickly (see e.g. [36]). We have in our model used the shorter parallel ballooning length for the electron heat conduction to include the fast decay of electron pressure in the SOL.

We have implemented the sheath condition in the model, linking the electron potential to the electron temperature in the SOL. This adds a significant sink for the poloidal velocity in the SOL. The sheath term is used to model connected and disconnected divertor conditions. In the connected condition we assume that both the mean and fluctuation part of the potential and electron temperature are coupled by the divertor sheath. Blobs being ejected into the SOL will in this case quickly lose momentum and will be left—somewhere in the middle of the SOL—to decay due to parallel loses. Under disconnected divertor conditions only the mean part of the potential and electron temperature are coupled by the divertor sheath. Strong and fast blobs are observed to be generated and they are mostly able to cross the SOL and enter the wall region, interacting with the first wall in a toroidal device. If the connection length becomes too long we also would assume that the fluctuating part of the potential and electron temperature cannot be coupled. Such a condition would resemble a disconnected sheath condition in our model.

The inclusion of drift waves in the edge region has proven to be quite important for the observed instabilities in this region. Generally, we did not observe pure drift waves with a phase relation between density and radial velocity close to $\pi/2$ in this region, but the drift wave term influences the length scale of the instability which essentially determines the size of the blobs being ejected into the SOL. This model is thus not significantly affected by the choice of aspect ratio of the numerical domain. The drift wave terms are very sensitive to the choice of parallel length scale, as $\alpha \sim L_\parallel^{-2}$. The parallel length scales are here chosen to be equal to the parallel ballooning length on open field lines, e.g. $L_\parallel = L_b = qR$. We observed blobs in the SOL with diameters of $1-2$ cm, in agreement with experimental observations (see e.g. [21]). A longer parallel length would result in larger blobs and for a length scale $L_{\parallel} < qR/4$ the gradients in the edge region appear to be stable.

By changing the collisionality we can change the ratio between perpendicular to parallel transport, defined by Λ (see equation (1)). The observed 'shoulder' formation in the density profile for increasing collisionality is in our case a result of local processes on the outboard midplane. We are, of course, aware that our parallel physics is simple, but we note that our results are in qualitative agreement with experimental observations in COMPASS, AUG and JET (see [18–20]) as well as with an older comparison between the numerical model, ESEL and Langmuir probe measurements at TCV (see [37, 38]). A simple way of changing the parameter Λ is to change the connection length L_c. Changing this parameter will only change the sheath term in the model. We can also in this case observe a shoulder in the density $\Lambda > 1$ (results not shown in this paper).

Finally, we have demonstrated that the energy transport across the SOL, and thus also the energy transport to the divertor region, is strongly intermittent and the nonlinear triple products in the energy flux terms are dominant. Energy transport is dominated by blob dynamics and we can observe a significant energy transport far into the SOL even in the case of flat density and temperature profiles.

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